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THE DETERMINANTS OF BANK INTEREST MARGINS: THEORY AND EMPIRICAL EVIDENCE

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I. Introduction

The recent experience of high and volatile interest rates has caused a number of financial intermediaries severe financial mananagement problems [17]. An area of particular concern has been the impact of rate volatility on bank interest margins or spreads. In a recent study, Ohlson *et al.* [18] argued that over time, commercial bank margins have become increasingly sensitive to interest rate volatility. A number of reasons could be put forward to explain this development. These include increasing reliance by banks on interest sensitive short-term liabilities, such as federal funds, while at the same time placing a greater emphasis on loans in the asset portfolio (many of which have contractually fixed or insensitive interest rates). As a result, interest-sensitive liabilities have tended to grow faster than interest-sensitive assets making it difficult for banks to "immunize" the effects of interest rate changes on interest margins.[1] Despite a growing concern by bankers and regulators over these trends, however, few theoretical models have been developed to explicitly analyze the determination of bank interest margins.[2]

To date, perhaps the best known models of bank behavior are the hedging hypothesis and those models developed from the microeconomics of the (banking) firm. The hedging hypothesis views the bank as seeking to match the maturities of assets and liabilities in order to avoid the reinvestment or refinancing risks which arise if assets are either too short or too long. Consequently, this type of model assumes that the major portfolio risk emanates from interest rate fluctuations.

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[1] The interest margin is defined here as the spread between the interest revenue on bank assets and interest expense on bank liabilities as a proportion of average bank assets--this margin is also called the banker's mark-up (see [18]).

[2] It might be noted that the importance of the relationship between interest volatility and bank portfolio behavior was recognized as early as Samuelson [21].

While this type of model appears to explain many aspects of actual bank portfolio behavior (see Dougall and Gaumnitz [6]), its principal weakness lies in the fact that its proponents usually fail, in any rigorous fashion, to tie hedging behavior to the underlying objective function of the decision maker even though an implicit assumption appears to be made that the bank hedges in order to minimize the risk of shareholders' wealth (see Michaelson and Goshay [16]).

The second group of models are those based on the microeconomics of the banking firm. These models extensively reviewed in articles by Pyle [20] and Baltensperger [1] exhibit a good deal of heterogeneity. Nevertheless, it is usually assumed that the bank seeks to maximize either expected profit (wealth) or the expected utility of profit (wealth). A paper by Pyle [19] provides a good example of the latter class of model and is directly related to the question of margin determination. Assuming that the bank is an expected utility of wealth maximizer, Pyle sought to determine the necessary and sufficient conditions for the existence of financial intermediation. Under some fairly simple assumptions about time and the asset opportunity set,[3] it was shown that if rates on deposits and loans were independent, intermediation would exist whenever a positive risk premium for loans and a negative premium for deposits were present.[4] No real attempt was made, however, to analyze the factors determining the size of these premiums (other than positive or negative) or how these premiums adjusted to changes in market interest rates and other variables.

The objective of this paper is to extend and integrate the hedging and expected utility approaches into an analysis of the determinants of bank margins. In developing the model we will initially abstract from many of the institutional constraints facing banks, as well as certain financial management problems such as overhead control and capital adequacy, which tend to be interdependent with bank margin determination (see Hempel and Yawitz [9]). In developing the model we also draw heavily on the growing literature on bid-ask prices for security market dealers; (see, for example, Ho and Stoll [10, 11] and Stoll [22,23].

In the model developed in Section II below, the bank is viewed as "a dealer" (essentially a demander of one type of deposit and supplier of one type of loan). In undertaking this function it faces a major type of uncertainty and, hence, cost. This cost occurs because the demand for bank loans and the supply of deposits are viewed as stochastic so that deposit supplies (inflows) tend to arrive at different times from loan demands (outflows). The stochastic behavior

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[3] A one-period model was selected with three assets: cash (the risk-free asset); loans; and deposits.

[4] More complex conditions were found when returns were assumed to be interdependent.

of loans and deposits results in the bank having to hold either a long or short position in the short-term money market. It is shown that the bank will demand a positive interest spread or fee as the price of providing immediacy of (depository and/or loan) service in face of the (transactions) uncertainty generated by asynchronous deposit supplies and loan demands. The model indicates that the optimal mark-up (sum of fees) for deposit and loan services will depend on four factors: (i) the degree of bank management risk aversion; (ii) the market structure in which the bank operates; (iii) the average size of bank transactions; and (iv) the variance of interest rates. It is also shown that even in a world of highly competitive banking markets, positive margins will tend to exist. That is, interest margins cannot disappear, under quite reasonable assumptions, as long as transactions uncertainty is present. Hence, we call the margin due to transactions uncertainty the pure spread.

In Section III an attempt is made to identify the size of this pure margin using quarterly income and balance sheet data for a sample of large commercial banks. In this section, account is taken of a number of imperfections and regulatory restrictions, not explicitly considered in our theoretical model, but which, nevertheless, are likely to impact on actual margins derived from balance sheet and income statement data. The factors directly considered are the probability of loan defaults, the opportunity cost of required reserves, and the cost of implicit interest payments on deposits in the form of service change remissions or subsidies. Using cross-section regression models it is shown that it is possible to estimate the pure transaction margin or spread for this group of banks for each quarter. Using these estimated spreads as a time series, the validity of the theoretical model is tested by examining the empirical relationship between the derived pure spreads and measures of interest rate volatility and market structure. Finally, in Section IV, the results are summarized and their implications for the theory of financial intermediation discussed.

II. The Model

The bank in the model is viewed as a dealer in the credit market acting as an intermediary between the demanders and suppliers of funds. To focus on the dealer role of a bank, we abstract from credit risk and assume that all deposits and loans are costlessly processed. Furthermore, we also assume a one-period decision model and that the bank maximizes the expected utility of terminal wealth (the decision period is assumed to be short).

The bank has three components to its wealth portfolio. The first component is its initial, or base wealth, Y, which is invested in a diversified portfolio. The second component is a net credit inventory I. It is assumed that deposits

(D) and loans (L) have the same maturity date, but that both mature after the end of the decision period. The difference in market values of deposits and loans defines the bank's credit inventory (I), where $I = L - D$. Because loans and deposits are assumed to mature after the end of the decision period, the credit inventory, I, will be subject to interest-rate risk (a point elaborated on below) The third component is the bank's short-term net cash or money market position, C, defined as the difference between money market loans C^L and borrowings C^B, both of which mature at the end of the decision period.[5] The bank can be either short or long in the money market. If, for example, it is short, the bank is financing some of its credit inventory by short-term money market debt.

In summary, the bank's wealth portfolio at the end of the decision period is given by:

$$W = \tilde{Y} + \tilde{I} + C$$

where

$$W = (1 + r_Y)Y_o + Y_o\tilde{z}_Y$$

$$I = (1 + r_I)I_o + I_o\tilde{z}_I$$

$$C = (1 + r)C_o$$

such that,

r_Y, r_I, r are the expected rates of return on base wealth, net credit inventory, and the net cash position, and $\tilde{z}_Y$ and $\tilde{z}_I$ are random variables impacting on rates of return. The distributions of $\tilde{z}_Y$ and $\tilde{z}_I$ are normal with $E(\tilde{z}_Y) = E(\tilde{z}_I) = 0$ and are stationary with respect to all the economic parameters of the model. The joint distributions of the returns are assumed to be bivariate normal.

The bank in the model is assumed to make new loans and accept deposits in a passive way. That is, the bank sets loan and deposit prices, P_L and P_D, and quantity is determined exogeneously.[6] These prices are defined as:

$$P_L = p - b$$

$$P_D = p + a$$

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[5]The cash portion may be thought of as the net position of the bank in the federal funds, interbank, and other short-term markets for temporary borrowings and loans.

[6]Hence, these deposits tend to exhibit the characteristics of time rather than demand deposits.

where p is the bank's opinion of the "true" price of the loan or deposit, and a and b are fees for the provision of immediacy of service. It should be noted that as defined here p, and therefore P_D and P_L, are prices, and as such are inversely related to deposit and loan rates. That is, a high deposit price implies a low deposit rate and vice-versa.

It is assumed that once the loan and deposit prices P_D and P_L are set at the beginning of the period, they remain unchanged over the rest of the period. We also assume, without loss of generality, that there is at most one loan or deposit arrival in this period with the same transaction size, Q. (In the case of multiple arrivals of loans and deposits, Q would be the sum of all loan (and deposit) transactions.

The probability of a new deposit supply (λ_a) and a new loan demand (λ_b) arriving at the bank depends on the respective sizes of the two fees a and b. For example, by raising b, the price of loans, P_L falls (rates on loans rise) and new loan demand is discouraged. By raising a, the price of deposits, P_D rises (deposit rates fall) so that new deposits are discouraged. Clearly, by manipulating the size of the fees a and b, and, therefore, the price or interest spread a + b, the bank can influence the probability of loan and deposit arrivals.

In this paradigm, because of the (assumed) long-term maturity of deposits and loans and the uncertainty over transaction arrivals, the bank will face an interest-rate risk whenever it holds an unmatched portfolio of deposits and loans at the end of the decision period and the short-term rate of interest (r) changes. That is, suppose a deposit is made at the bank at some long-term rate r_I. If this deposit arrives at a different instant in time from a new loan demand, the bank will have to temporarily invest the funds in the money market at the short-term risk-free rate r. In so doing the bank faces a reinvestment risk at the end of the decision period should the short-rate fall. Elsewhere, Cox, Ingersoll and Ross [4] have called this the basis risk of the portfolio. Similarly, if the demand for a new loan is met by the bank without a contemporaneous inflow of deposits, the bank would have to resort to short-term borrowings in the money market at rate r to fund the loan. In other words, the bank's net credit inventory (I = L - D) would increase and its net money market position, C, decrease. In this case the bank is subjecting itself to a refinancing risk at the end of the decision period should the short-term rate, r, rise. The larger the net credit inventory, I, in absolute terms, the greater the interest-rate risk the bank faces.

Hence, the bank's decision problem in the face of these transaction and interest-rate risks is to determine the optimal, expected utility-maximizing, deposit and loan rates or deposit-loan interest spread(s) (where s = a + b using our previous notation). The expected utility of wealth at the end of the period is given by

$$(1) \qquad EU(\tilde{W}) = U(W_0) + U'(W_0)r_w W_0 + \frac{1}{2}U''(W_0)(\sigma_I^2 I_0^2 + 2\sigma_{IY} I_0 Y_0 + \sigma_Y^2 Y_0^2)$$

where

$$r_W = r_Y \frac{Y_0}{W_0} + r_I \frac{I_0}{W_0} + r \frac{C_0}{W_0} .$$

For the case when a new deposit transaction is made, the bank's credit inventory is $I_0 - Q$, where Q is the size of the transaction and its cash or short-term money market position is $C_0 + Q + Qa$, i.e., its initial cash position plus the deposit inflow plus the fee charged for depository immediacy times the size of the deposit. Substituting into equation (1), we have

$$(2) \qquad \begin{aligned} EU(\tilde{W} \mid \text{one deposit transaction}) = \; & U'(W_0)\; aQ + \frac{1}{2}U''(W_0)(\sigma_I^2 Q^2 + 2\sigma_I^2 Q I_0) + U(\\ & + U'(W_0) r_W W_0 + \frac{1}{2}U''(W_0)(\sigma_I^2 I_0^2 + 2\sigma_{IY} I_0 Y_0 + \end{aligned}$$

assuming that the second order of deposit fees are negligible and that $r_I = r - \frac{1}{2}\frac{U''}{U'}\sigma_{IY}$ (see Stoll [22]).

Similarly when a loan transaction is made, the bank's inventory is $I_0 + Q$ and its cash position is $C_0 - Q + Qb$. Substituting into equation (1) we have:

$$(3) \qquad \begin{aligned} EU(\hat{W} \mid \text{one loan transaction}) = \; & U'(W_0) bQ + \frac{1}{2}U''(W_0)(\sigma_I^2 Q^2 - 2\sigma_I^2 QI) + U(W_0) \\ & + U'(W_0) r_W W_0 + \frac{1}{2}U''(W_0)(\sigma_I^2 I_0^2 + 2\sigma_{IY} I_0 Y_0 + \sigma_Y^2 \end{aligned}$$

Since the probability of a deposit or loan transaction is given by λ_a and λ_b respectively, it follows that the expected utility of wealth for fees a and b is given by

$$(4) \qquad EU(\tilde{W} \mid a,b) = \lambda_a EU(\tilde{W} \mid \text{one deposit transaction}) + \lambda_b EU(\tilde{W} \mid \text{one loan transactio}$$

Assuming symmetric and linear supply of deposit and demand for loan functions, we have

$$\lambda_a = \alpha - \beta_a \tag{5}$$

$$\lambda_b = \alpha - \beta_b \tag{6}$$

Since the fees a*, b* are set such that the expected utility of wealth is maximized, we require

$$\frac{\partial EU}{\partial a}(\tilde{W} \mid a^*,b^*) = 0 \tag{7}$$

and

$$\frac{\partial EU}{\partial b}(\tilde{W} \mid a^*,b^*) = 0. \tag{8}$$

It follows from equations (2), (3), (4), and (5) that equation (7) becomes

$$- \beta[U'(W_0)aQ + \frac{1}{2}U''(W_0)\sigma_I^2(Q^2 + QI)] + (\alpha - \beta_a)U'(W_0)Q = 0 \tag{9}$$

and equation (8) becomes

$$- \beta[U'(W_0)bQ + \frac{1}{2}U''(W_0)\sigma_I^2(Q^2 - QI)] + (\alpha - \beta_b)U''(W_0)Q = 0. \tag{10}$$

Simplifying (9) and (10) and rearranging we have

$$s = a + b = \alpha/\beta - \frac{1}{2}\frac{U''}{U'}\sigma_I^2 Q. \tag{11}$$

If we define the coefficient of absolute risk aversion $R = -\frac{U''}{U'}$, then

$$s = a + b = \alpha/\beta + \frac{1}{2}R\sigma_I^2 Q. \tag{11'}$$

The first term, α/β, measures the bank's risk neutral spread,[7] and is the ratio of the intercept (α) and the slope (β) of the symmetric deposit and loan arrival functions of the bank. A large α and a small β will result in a large α/β ratio and, hence, spread (s). That is, if a bank faces relatively inelastic demand and supply functions in the markets in which it operates, it may be able to exercise monopoly power (and earn a producer's rent) by demanding a greater spread than it could get if banking markets were competitive (low α/β ratio).

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[7]This would be the spread if the coefficient of absolute risk aversion, R, was zero and the bank was, therefore, an expected wealth maximizer.

Consequently, the ratio α/β provides some measure of the producer's surplus or monopoly rent element in bank spreads or margins. The second term is a first-order risk-adjustment term and depends on three factors: (i) R, the bank management's coefficient of absolute risk aversion; (ii) Q, the size of bank transactions; and (iii) σ_I^2, the "instantaneous" variance of the interest rate on deposi and loans. Note that the second term implies that, *ceteris paribus,* the greater the degree of risk-aversion, the larger the size of transactions and the greater the variance of interest rates, the larger bank margins are. This spread equation has an important implication for the microfoundations of financial intermedia tion since it implies that even if banking markets are highly competitive, as lon as bank management is risk-averse and faces the type of transactions uncertainty defined in this paper, positive bank margins will exist as the price of providing deposit-loan immediacy.

A further interesting implication of equation (11') is that the size of the spread itself is independent of the size of the credit inventory (I). That is, the size of I does not enter as a determining variable in equation (11'); instead it only affects the adjustment of the spread (d) relative to the true price p; that is, by rearranging (9) and (10), we have

$$d = b - a = - 1/2R\sigma_I^2 I. \tag{12}$$

In other words, suppose deposit inflows are greater than loan demand so that the bank has to increase its short-term money market investments. Equations (11') and (12) imply that the bank reacts by adjusting its fee, a, upwards to discourage additional deposits and its fee, b, downwards to encourage extra loans so that its fee spread, a + b, remains the same. Similarly, if loan demand was greater than deposit supply, b would be raised and a lowered. Again the spread, a + b, will remain unchanged although its location relative to the true price p will change. By following such pricing behavior, the bank is trying to hedge or match the duration of its assets and liabilities. This implication can also be seen by noting, from equation (12), that when I = 0 (and the bank is fully hedged then d = 0, i.e., no fee adjustment takes place.

In the next section we attempt to empirically estimate the size of the pure spread, s, derived from our theoretical model developed above. The test procedure followed and the results are discussed in Section III below.

III. Empirical Tests

The data employed in the estimation of the pure margin or spread, s, were comprehensive income and balance-sheet data on major U.S. banks published by

Keefe, Bruyette and Woods, Inc. These data have been published for over 100 major banks for 13 quarters (1976-IV to 1979-IV).[8] Among the income and balance sheet ratios published is each bank's interest margin (i.e., interest income-interest expense/earning assets). In order to estimate the pure margin or spread, however, certain market and institutional imperfections have to be taken into account. These factors, while not explicitly considered in Section II, will, nevertheless, impact on actual margins as reflected in published accounting data.

The three factors or imperfections considered here, while not totally inclusive, were felt to be reasonably amenable to measurement. The first imperfection was bank payments of implicit interest (instead of, or as well as, explicit interest) on deposits, through service charge remissions and other types of depositor subsidy, due to regulatory restrictions on explicit interest payments (see [12]). These implicit interest payments should be regarded as an extra interest expense and are likely to be reflected in actual bank margins or mark-ups.[9] In a world without rate regulation, where banks can pay a competitive rate on demand and time deposits and charge a price equal to marginal cost for services supplied, published interest expense would be more representative of true interest expense than under the current regulatory structure. The second imperfection, likely to impact on bank margins, is the bank's opportunity cost of holding required reserves. The existence of noninterest-bearing reserve requirements increases the economic cost of funds over and above the published interest expense.[10] This additional cost factor will depend on the size of reserve requirements as well as on the opportunity cost of holding reserves (the short-term, risk-free rate). The third imperfection is the default risk on loans. If there is some probability of borrowers defaulting on loans, then an additional premium is likely to be added to the rate on loans and be reflected in the actual margin. The greater the probability of loan change-offs, and, therefore, potential loss of capital and interest, the greater the default premium likely to be demanded. Consequently, at any moment in time, it is hypothesized that actual bank margins (M) will comprise a pure spread (s) due to underlying transactions uncertainty, plus mark-ups for implicit interest expense (IR), the opportunity cost of required reserves (OR), and default premiums on loans (DP). All other market imperfections and regulatory restrictions impacting

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[8] Only 53 banks were found to comprehensively cover the full data period and these were used as the sample.

[9] The Federal Reserve Board [2] has recently estimated that the implicit interest payments on consumer demand deposits alone averaged 4.43 percent.

[10] Mason [15, Ch. 10] provides an example to demonstrate this.

on M are reflected in the portmanteau, or residual, variable U.[11] Thus,

$$M = f(s(\cdot), IR, OR, DP, U) \tag{13}$$

where the s (•) function defines the factors impacting on the pure spread derived in equation (11') in Section II. From equation (11') it can be inferred that even if a group of banks have heterogeneous net credit inventories (I) at any moment in time, i.e., $(L_i-D_i) \neq (L_j-D_j)$, as long as they share similar attitudes to risk (R), size of transactions (Q), market structure (α/β), and interest rate volatility (σ^2), their pure interest spreads or margins will be the same, i.e., $s_1 = s_2 = \ldots = s_n$.[12] Hence, we can define a cross-sectional regression equation for the group of banks of the form:

$$M_i = \delta_0 + \delta_1 IR_i + \delta_2 OR_i + \delta_3 DP_i + U_i \tag{14}$$

where δ_0, or the regression constant, provides an estimate of the pure spread component for the banks in the sample. The results of the cross-sectional regressions on 53 banks for each quarter are presented in Table 1 along with definitions of the variables employed. As can be seen, the pure spread estimate, δ_0, and the implicit interest rate variable, (IR), are statistically significant at the 5 percent level in all regressions, while the coefficients on the opportunity cost of reserves (OR), and default premiums, (DP), are generally small and insignificantly different from zero. The $\bar{R}^2$s are also reasonably high, varying between .51 and .79. Hence, these results imply that the main determinants of the size of actual bank margins were transactions uncertainty and mark-ups to cover implicit interest payments to depositors. Over time the estimated pure spread, ($\hat{\delta}_0$), varied from a low of 2.21 percent to a high of 3.22 percent with an average value of 2.637 percent. This compares to a low in actual margins of 4.44 percent, a high of 4.91 percent, and an average value of 4.68 percent. That is, the pure spread as measured here, accounted for 56 percent of the size of actual bank margins for the banks in our sample.

It can also be seen from Table 1 that the estimated pure spread, ($\hat{\delta}_0$), varied

[11] This might include such factors as the regulatory expense due to FDIC insurance premiums (see Dufey and Giddy [7]) and the use of compensating balances as a means of keeping published loan rates below the true loan rate (see Burns [3]). As data were not available on compensating balance requirements for the banks in the sample, the compensating balance component was not explicitly analyzed as a separate variable and, like FDIC insruance costs, was subsumed in U_i.

[12] This assumption is subjected to greater scrutiny later in the paper.

TABLE 1

THE CROSS-SECTIONAL RELATIONSHIPS BETWEEN THE INTEREST MARGIN, THE PURE SPREAD, AND OTHER DETERMINING VARIABLES

$$M_i = \delta_0 + \delta_1 IR_i + \delta_2 OR_i + \delta_3 DP_i + U_i$$

	δ_0	δ_1	δ_2	δ_3	$\bar{R}^2$
1976 - IV	3.2285*	2.6772*	- .5591	.2457	.5199
1977 - I	2.8358*	3.4491*	-1.0056	.3930	.6699
1977 - II	2.7365*	3.6509*	- .9833	.6879	.6559
1977 - III	2.6579*	4.0551*	.7827*	- .7256	.6829
1977 - IV	3.1401*	3.0675*	.4216	- .3893	.5277
1978 - I	2.7330*	3.7066*	.5302*	1.0649	.7360
1978 - II	2.6759*	4.1649*	.4753*	3.4054*	.7907
1978 - III	2.4470*	3.9736*	- .0178	.3637	.6783
1978 - IV	2.6554*	3.5981*	- .1415	1.1385	.5727
1979 - I	2.2960*	4.0928*	- .0403	1.0772	.6537
1979 - II	2.3137*	4.1226*	.1377	1.9601	.6896
1979 - III	2.3564*	4.0703*	.1470	1.8258	.6874
1979 - IV	2.2112*	4.1612*	.0599	1.5861	.6707

Notes: *Significantly different from zero at the 5% level.

(I) M_i = (interest income - interest expense/total earning assets)

(II) IR_i = (total noninterest expense - total noninterest revenue/total earning assets)

(III) OR_i = (total noninterest-bearing reserve assets/total earning assets) X the average treasury bill rate

(IV) DP_i = (net loan chargeoffs/total earning assets) over time from quarter to quarter. Our model in Section II (equation (11')) suggested four possible reasons for this. If it is assumed, however, as seems reasonable, that the coefficient of risk aversion, (R), the size of transactions, (Q), and market structure, (α/β), change relatively slowly over time compared to interest rate changes, then we might expect the quarterly time series behavior of the estimated pure spreads, (δ_t), to be largely dependent on changes in the volatility or variability of interest rates, (σ^2). That is, we can specify

$$\delta_t = \gamma_0 + \gamma_1 \sigma_t^2 + \varepsilon_t \tag{15}$$

where from the theoretical model, equation (11'), we have

$$\gamma_0 = \alpha/\beta \text{ and } \gamma_1 = 1/2RQ$$

and σ_t is some measure of the "instantaneous" variance of interest rates. There might appear to be two problems in estimating an equation such as (15). The first is in defining "the interest rate"; the second is in measuring the instantaneous variance of that rate for each time period. The first, in fact, presents few problems since by using a whole structure of rates, we can test the sensitivity of δ_t to the variability in rates for a number of maturities. In so doing we are implicitly measuring the time horizon or asset (and liability) duration of concern to bank management. For example, a highly significant coefficient on the σ_t^2 variable for a particular maturity implies that this is the portfolio duration of major concern to the bank. The second problem is in the calculation of σ_t^2 itself. This presents a very difficult measurement exercise because of the discrete nature of the bond interest time-series that are available.[13] Dothan [5] has attempted to deal with this problem by using available data on very short-term rates to generate the whole term structure. His method, however, employs assumptions that amount to making tests with the derived interest series joint hypotheses. For example, Dothan assumes both that the short-rate drives the term structure and that this rate follows a particular stochastic process over time. Rather than using these rather rigid transformations, and given the constraints imposed by data availability, we used the quarterly variance of each (weekly) interest series, pre-

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[13] For example, the only consistent, publicly available, weekly time series on government bond yields of different maturities are published by Salomon Brothers. Consistent daily series are not publicly available.

whitened by an optimal filter, as empirical surrogates for σ_t^2.[14,15] The regression tests of equation (15) using the variances of pre-whitened interest rates as surrogates for the σ_t^2 of five different maturities, are presented in Table 2 below.

TABLE 2

THE RELATIONSHIP BETWEEN THE PURE SPREAD AND THE VARIANCE OF DIFFERENT INTEREST RATES OVER TIME

Model: $\delta_t = \gamma_0 + \gamma_1 \sigma_t^2 + \varepsilon_t$

Rate of Interest	γ_0	γ_1	R^2	D.W.
3 month	1.583* (21.333)	-.002 (0.248)	.01	2.03
1 year	2.229* (36.622)	.0568* (3.097)	.52	2.10
2 year	1.849* (25.908)	.012 (1.020)	.10	1.88
3 year	1.651* (24.889)	.004 (1.080)	.11	1.88
5 year	1.586* (23.931)	.0012 (1.058)	.11	1.86

Notes: t - ratios in parentheses,

* indicates significant at the 5% level.

Where appropriate the Cochrane-Orcutt iterative technique was used to eliminate any significant first-order autocorrelation in the residuals.

As can be seen, the only γ_1 (slope coefficient) statistically significant at the 5 percent level is that on the variance of one-year bond rates. Indeed, the one-year variability equation is far better determined (in terms of R^2) than the other four equations (3 months, 2 years, 3 years, and 5 years). Hence, it

[14]The interest rate time series used were weekly series on treasury bonds of different maturities published by Salomon Brothers.

[15]This procedure was followed because of the theoretical requirement that the underlying time series generating σ^2 should follow a random walk. Whereas raw stock returns tend to follow such a pattern, interest rates do not and, hence, the need to use a filter to pre-whiten the interest rate series.

appears, over our admittedly short data period, that the pure spread positively responded to interest rate variability and that the duration of the asset (liability) portfolio was equal to one year. It might also be noted here that this "duration" result is quite close to that found by Flannery ([8, p. 21]), who, using a very different methodology based on speeds of asset and liability adjustment, estimated that average bank asset and liability maturities were 1.5 and 1.7 years, respectively. Our results, albeit for a shorter time period, found an average maturity of approximately one year. Of course, the absence of a consistent published time series on one and one-half year bond rates prevents a test of equivalence of results. Nevertheless, given the differences between Flannery's and our approaches, the findings are surprisingly close.[16]

So far we have assumed that all the banks in our sample are homogeneous, at any moment in time, in terms of the variables impacting on the pure spread. One implication of the model that deserves analysis, however, is the extent to which the pure spread (s) is dependent on differences in the structure of the deposit and loan markets in which banks operate. It was argued in Section II that structure could be inferred (at least within the framework of the model) from the slope or price elasticities of the deposit supply and loan demand functions. The lower these elasticities, the easier it is for a bank to generate profits by raising its fees, thereby widening its interest spread. As our sample of banks included both large money center banks (such as Chase Manhattan) as well as a number of smaller, more regionally oriented banks, such as the Bank of Virginia, it was decided to split the sample into the largest and smallest banks according to asset size.[17] It was our *a priori* belief that the larger banks tend to be more competitive, both nationally and internationally, and, therefore, operate in a more competitive market framework than the smaller banks, who may be more localized and better able to exploit regional monopoly positions. Indeed, there is some evidence that interest margins are narrower for larger banks.[18] The next step was to examine whether this observation could be explained by larger banks having narrower pure transaction spreads. Table 3 below presents the estimated δ_0's

[16] It might also be noted that the results in the one-year regression imply that $\alpha/\beta = \hat{\gamma}_0 = 2.229$. That is, the average producer's rent element for this group of banks as a whole was 2.229 percent.

[17] Large banks were defined as those banks whose assets in every quarter were above the mean of the sample, and small banks were defined as those whose asset size fell below the mean for every quarter. After separation, 22 banks were placed in the large group and 22 in the smallest group with nine banks excluded.

[18] See, for example, Flannery [8].

from cross-sectional regressions of equation (14) for the small and large bank subsamples in each quarter. The means of each δ_0 series are also tabulated.

TABLE 3

ESTIMATED δ_0'S FOR THE SMALL AND LARGE BANK SUBSAMPLES

	δ_0 Small Banks	δ_0 Large Banks
1976 - IV	4.0021	3.1140
1977 - I	3.3631	3.0801
1977 - II	3.2727	3.1868
1977 - III	3.2593	3.1218
1977 - IV	3.8062	3.5322
1978 - I	3.4227	3.0817
1978 - II	3.3334	2.9485
1978 - III	3.4629	2.4883
1978 - IV	3.2663	3.1738
1979 - I	3.0646	2.8112
1979 - II	2.9658	2.7253
1979 - III	2.9875	2.7872
1979 - IV	3.1082	2.4772
MEAN	3.3319	2.9637

As can be seen, the mean of large bank δ_0's is 2.9639 percent compared to a mean of 3.3319 percent for small banks, or a difference in pure spreads of .368 percent. Moreover, a t-test reveals that t = 4.47, i.e., the means are statistically different at the 5 percent level. Hence, it appears that the largest banks in the sample had a slightly smaller (pure) margin than the smallest banks over the period of analysis.

It is interesting to examine whether this difference was due to market structure factors (α/β) alone or also due to differences in risk aversion parameters (R) and transaction sizes (Q). Consequently, equation (15) was rerun for both small and large bank subgroups to test the null hypotheses

$$\gamma_0 \text{ small} = \gamma_0 \text{ large and } \gamma_1 \text{ small} = \gamma_1 \text{ large.}$$

The results of the regressions for subgroup δ_t's on σ_t^2's are shown in Table 4.

Looking first at the γ_1 coefficients, it can be seen that the only small bank γ_1 coefficient that is statistically different from zero at the 10 percent level is that in the one-year maturity regression. For large banks, the γ_1 coefficient in the one-year maturity regression is also statistically significant at the 10 percent level. For large banks, however, both the three- and five-year σ_t^2 variables have positive and significant γ_1 coefficients,[19] although they are smaller in absolute value than in the one-year regressions.

To test the null hypotheses of equality, for smaller and larger bank γ_0's and γ_1's, stacked regression tests were conducted following Maddala [13, p. 112]. These involved running, both constrained and unconstrained, equation (15) and then conducting F tests on the sum of squared residuals from each equation.[20]

Using the one-year maturity series to measure σ_t^2, the null hypothesis that $\gamma_0^S = \gamma_0^L$ was rejected at the 10 percent level (F = 3.12) while the null hypothesis that $\gamma^S = \gamma^L$ could not be rejected at any meaningful level (F = .021). Consequently, there is some evidence, admittedly fairly weak, that the relatively small difference in the average of the pure spreads for the larger and smaller banks in our sample was due to differences in market structure rather than risk aversion and transactions size. Clearly this is an area that needs further investigation with a larger sample of banks and also over a longer period of time.

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[19] At the 5 percent level.

[20] In matrix form we run the unconstrained stacked regression (1)

(1)
$$\begin{bmatrix} \delta^S \\ \hline \delta^L \end{bmatrix} = \left[\begin{array}{c|c|c|c} \sigma^2 & 0 & 1 & 0 \\ \hline 0 & \sigma^2 & 0 & 1 \end{array}\right] \begin{bmatrix} \gamma_1^S \\ \gamma_1^L \\ \gamma_0^S \\ \gamma_0^L \end{bmatrix} + \begin{bmatrix} \varepsilon_S \\ \hline \varepsilon_L \end{bmatrix} .$$

followed by the constrained regression where $\gamma_0^S = \gamma_0^L$

(2)
$$\begin{bmatrix} \delta^S \\ \hline \delta^L \end{bmatrix} = \left[\begin{array}{c|c|c} \sigma^2 & 0 & 1 \\ \hline 0 & \sigma^2 & 1 \end{array}\right] \begin{bmatrix} \gamma_1^S \\ \gamma_1^L \\ \gamma_0 \end{bmatrix} + \begin{bmatrix} \varepsilon_S \\ \hline \varepsilon_L \end{bmatrix}$$

and then calculate

(3)
$$\frac{(SSR_2 - SSR_1)\ /2}{SSR_1\ /\ n-k-1} \sim F(2,\ n-k-1)$$

where SSR_2 is the sum of squared residuals in the constrained regression and SSR_1 the sum of squared residuals in the unconstrained regression and the ratio is F distributed. A similar procedure is then followed to test $\gamma_1^S = \gamma_1^L$.

TABLE 4

RELATIONSHIP BETWEEN δ_t AND σ_t^2 FOR SMALL AND LARGE BANKS

	Small Banks				Large Banks			
	γ_0	γ_1	R^2	D.W.	γ_0	γ_1	R^2	D.W.
3 Months	2.2956*	- .00126	.0034	1.501	2.9852*	.0034	.0116	1.608
	(29.901)	(-.175)			(30.222)	(.343)		
1 Year	2.387*	.0404**	.296	1.480	2.9491*	.03604**	.1736	1.852
	(36.346)	(1.945)			(35.190)	(1.779)		
2 Years	1.867*	.0115	.0988	1.812	2.9805*	.0104	.0502	1.676
	(25.810)	(.936)			(34.468)	(.727)		
3 Years	2.365*	- .0028	.0417	1.377	2.1896*	.0111*	.4092	1.899
	(33.929)	(-.626)			(31.746)	(2.497)		
5 Years	2.3556*	- .0009	.0662	1.426	1.9708*	.0032*	.4651	1.819
	(34.191)	(-.798)			(29.951)	(2.798)		

Notes: *Significant at the 5% level.

**Significant at the 10% level.

IV. Summary and Conclusion

This paper has developed a model of bank margins or spreads in which the bank is viewed as a risk-averse dealer. It was demonstrated that an interest spread or margin would always exist, and that this was the result of transactions uncertainty faced by the bank. Moreover, it was shown that this pure spread depended on four factors: the degree of managerial risk aversion; the size of transactions undertaken by the bank; bank market structure; and the variance of interest rates. The model implied that liability and asset structures had to be analyzed together since they were directly interrelated through transactions uncertainty. It was shown that because of this transactions uncertainty, hedging behavior was perfectly rational within an expected utility maximizing framework. Extending the model from a structure with one kind of loan and deposit to loans and deposits with many maturities should lead to further interesting insights into margin determination especially as "portfolio" effects may become apparent.

In the empirical section, tests were undertaken using data from a sample of U.S. commercial banks. It was shown, given certain reasonable assumptions, that the pure spread could be empirically identified. It was found that this spread was positively and significantly related to the variance in the rate on bonds as predicted by the theoretical model. It was also inferred that the asset (liability) maturity, or duration, of major concern to this group of banks was one year. When the sample was split into two subgroups (measured by asset size), it was found that the smaller banks had an average transactions spread of approximately one-third of one percent more than the larger banks. While this difference was small, it appeared to be statistically significant.

Statistical tests were then conducted to see which factors explained this difference. It was tentatively concluded that the difference was largely due to market structure factors which allowed the smaller banks to earn some additional "producer's" rent or profit.

Finally, this study suggests that the size of bank spreads or margins is directly amenable to theoretical and empirical modeling. Consequently models similar to the one developed in this paper may provide useful financial management tools for banks in the face of increased uncertainties over the future course of interest rates.

REFERENCES

[1] Baltensperger, E. "Alternative Approaches to the Theory of the Banking Firm." *Journal of Monetary Economics,* Vol. 6 (1980), pp. 1-37.

[2] Board of Governors of the Federal Reserve System. "The Impact of the Payment of Interest on Demand Deposits." Staff study (1977).

[3] Burns, J. E. "Compensating Balance Requirements Integral to Bank Lending." Federal Reserve Bank of Dallas, *Business Review* (1972), pp. 1-8.

[4] Cox, J. C.; J. C. Ingersoll; and S. A. Ross. "A Theory of the Term Structure of Interest Rates." Unpublished working paper (1977).

[5] Dothan, L. V. "On the Term Structure of Interest Rates." *Journal of Financial Economics,* Vol. 6 (1978), pp. 59-69.

[6] Dougall, H., and J. E. Gaumnitz. *Capital Markets and Institutions.* Englewood Cliffs, N.J.: Prentice-Hall (1975).

[7] Dufey, G., and I. H. Giddy. *The International Money Market.* Englewood Cliffs, N.J.: Prentice-Hall (1978).

[8] Flannery, M. J. "Market Interest Rates and Commercial Bank Profitability: An Empirical Investigation." Unpublished working paper, University of Pennsylvania (1980).

[9] Hempel, G., and J. Yawitz. *Financial Management of Financial Institutions.* Englewood Cliffs, N.J.: Prentice-Hall (1974).

[10] Ho, T. and H. Stoll. "On Dealers' Markets under Competition." *Journal of Finance,* Vol. 35, Papers and Proceedings (1980).

[11] __________. "Optimal Dealer Pricing under Transactions Return Uncertainty." *Journal of Financial Economics,* forthcoming.

[12] Klein, M. A. "The Implicit Deposit Rate Concept: Issues and Applications." Federal Reserve Bank of Richmond, *Economic Review September/October* (1978), pp. 3-12.

[13] Maddala, G. S. *Econometrics.* New York: McGraw Hill (1977).

[14] Maisel, S. J., and R. Jacobsen. "Interest Rate Changes and Commercial Bank Revenue and Costs." *Journal of Financial and Quantitative Analysis* (November 1978), pp. 687-700.

[15] Mason, J. M. *Financial Management of Commercial Banks.* New York: Warren, Gorman and Lamont (1979).

[16] Michaelson, J. B., and R. C. Goshay. "Portfolio Selection in Financial Intermediaries: A New Approach." *Journal of Financial and Quantitative Analysis* (June 1967), pp. 166-200.

[17] *New York Times,* Business Day (July 29, 1980), p. D1.

[18] Ohlson, R. L.; D. G. Simonson; S. R. Reber; and G. H. Hempel. "Management of Bank Interest Margins in the 1980's." *The Magazine of Bank Administration* (March 1980), pp. 30-46.

[19] Pyle, D. H. "On the Theory of Financial Intermediation." *Journal of Finance,* Vol. 28 (June 1971), pp. 737-747.

[20] __________. "Descriptive Theories of Financial Institutions under Uncertainty *Journal of Financial and Quantitative Analysis* (December 1972), pp. 2009-2031.

[21] Samuelson, P. "The Effects of Interest Rate Increases on the Banking System." *American Economic Review* (March 1945).

[22] Stoll, H. R. "The Supply of Dealer Services in Security Markets." *Journal of Finance* (September 1978), pp. 1133-1533.

[23] __________. "The Pricing of Security Dealer Services: An Empirical Study of NASDQ Stocks." *Journal of Finance* (September 1978), pp. 1153-1173.